

WANG YAO

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EDUCATION

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|-----------------|---|
| 09/2023-10/2024 | Faculty of Construction and Environment, HongKong Polytechnic University <ul style="list-style-type: none">➤ Master of Science in Sustainable Urban Development➤ Final GPA 3.51/4.3 |
| 09/2017-06/2022 | College of Civil Engineering and Architecture, Hebei University (HBU) <ul style="list-style-type: none">➤ Bachelor of Engineering in Architecture➤ A 5-year undergraduate programme with a final GPA of 3.88/5 |

RESEARCH INTERESTS

XR, Culture Heritage, Virtual Environmental

PUBLICATIONS & RESEARCH PROJECTS

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- Yufei Lu; Ye Jia; Guang Chen; **Yao Wang**; Peter H. F. Ng; Laura Zhou; Qing Li; Chen Li
Towards Effective Collaborative Learning in Edu-Metaverse: A Study on Learners' Anxiety, Perception, and Behaviour. **Learning Technologies and Systems (ICWL 2024)**
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| 10/2024-03/2025 | Breathing Scenery: Impact of Respiration Interaction in Nature Therapy <ul style="list-style-type: none">➤ The study contributes a novel framework integrating implicit/explicit cues and bio-responsive interaction in VR to significantly enhance box-breathing training efficacy, advancing immersive respiratory intervention design.➤ Unity development, Environment design, Experimental design |
| 11/2024-03/2025 | UI Design for a Smart Gardening System for Dementia <ul style="list-style-type: none">➤ Therapyuser interfaces for farming therapy applications, focusing on accessible design principles. The project emphasized creating calming, intuitive interfaces that support mental health and wellness through digital guidance of gardening experiences.➤ UI Design, Figma |
| 09/2020-06/2021 | Immersive Participation in Ancient Villages' Development in Context of Rural Revitalization: exemplified by tech-stock-based cooperation development in Yuanqu Village of Handan <ul style="list-style-type: none">➤ This project aims to revitalize Yuanqu Village in Henan as the "First Ancient Village East of Taihang Mountains" through a design-for-equity model, where a design firm contributes plans as shares and commits to holistic development, achieving tripartite success in heritage conservation, tourism revival, and commercial operations.➤ Played a responsible role in brainstorming, progress management and report writing➤ Gained approval by 2020 Hebei Province College Student Innovation & Entrepreneur Training Project |
| 09/2019-06/2020 | Development of An App Called "Yigeng Murals": the digital regeneration and activation plan of Taihang Murals <ul style="list-style-type: none">➤ Endeavored to help in spreading local culture and protecting Taihang |

Murals by achieving artistic activation of those murals and integrating them into modern life, with adoption of digital technologies

- Assisted project leader in ideas generation, task division, internal communication and report drafting
- Obtained a 3rd Prize in the “Challenge Cup” Hebei Province College Student Entrepreneurship Contest

09/2018-08/2019

Influence and Enlightenment of How Urban-Rural Economic and Cultural Differences Affect Formation of Middle School Students' Consumption View: A Case Study in Baoding and Cangzhou in Hebei

- Explored whether and how values and consumption views of middle school students are affected by the urban-rural economic gap while comparing their living environment and economic, cultural, educational and family background
- Won a Provincial 3rd Prize for "Research Hebei Project"

WORK & INTERNSHIP

10/2024 to 03/2025

Virtual Intern, Australian Research Centre for Interactive and Virtual Environments (IVE)

- Collaborate on the development of the virtual environment for VR breathing training projects, including programming and interactive design.
- Refining the experimental proposal, conducting experiments, collecting data, and analyzing results

07/2024 to 10/2025

Research Assistant, Department of Computing, HKPolyU

- Responsible for modeling and scene construction of virtual environments in projects.
- Provide and design prototypes for the UI design portion of the application build.

Literature collection of human-computer interaction, as well as the development of research ideas.

05/2023-08/2023

Intern, Shenzhen Urban Planning & Land Resource Research Center

- Participated in some important planning or policies for Shenzhen, working as an assistant Urban and Rural Planner.

Trained practical abilities of project design, team cooperation, and information gathering and organization

06/2022-08/2022

Game Art Designer, Beijing Bilibili Yuanli Xingju Network Tech Co., Ltd

- Responsible for game module integration, map building and iteration, and model-related data processing

Enhanced understanding of stylized scene construction, ability of data analytics and overall planning, and spatial processing and 3D modelling skills

PROFICIENCY & SKILLS

Language

- IELTS overall 7.0

Fine Art

- Aquarelle, Mark Painting, sketch

Software

- Unity, Blender, Figma, Photoshop, Rhino, Maya, Stata, AutoCAD

Photography

- A signed contributor to tuchong.com, an online community of photographers

Writing Expression	➤ Posted poems of over 10,000 words on Jianshu.com, an online platform for literature creators
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ACTIVITIES

09/2019	Surveyor, Ancient Buildings Mapping & Protection in Yuanqu Village, Handan ➤ Surveyed and mapped the villages dating back to the Ming and Qing Dynasties ➤ Sorted out and summarized the data collected
10/2019	Surveyor, Mapping and Research Project related to Baoding Industrial Heritage Involved in mapping and data collection of the old plant site
09/2017-06/2020	President, Taihang Local Culture Research Society Led peers to carry out various activities, e.g. field trips and photo exhibitions, to investigate or promote local cultures

AWARDS

Provincial-Level Grant, Provincial Third Prize	<i>National College Student Innovation & Entrepreneurship Training Program "Chuang Qingchun" National College Student Innovation & Entrepreneurship Competition</i>
Second Prize	<i>University-Level "Literary Poetry Gala" Poetry Contest</i>
Merit-based Scholarships	<i>Second Prize (twice) & Third Prize, awarded annually for three consecutive years</i>
Commendation Award	<i>PolyU Metaverse Contest 2023/24</i>